

Visual Behaviour in Searching Information: An Eye Tracking Preliminary Study

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Abstract—Human Computer Interaction (HCI) is a discipline that has progressed to establish a smooth understanding between humans and computers for effective visual searchability, accessibility and usability. With a lot of advancements in the search engines, only a few studies examined how the results are viewed from the perspective of end users from different regions. The main aim of this research is to investigate and explore whether there are cultural variations in visual search behaviour and eye movements on Search Engine Results Page (SERP) between two different cultures (Arabic and English), using the eye-tracking method in order to provide insights for global implementation of search engines. This research also aims to explore to what extent users with different language and cultural background differ in their visual search behaviour, performance and satisfaction when they are searching SERP with different layout and tasks. As a preliminary work, a controlled eye-tracking experiment was conducted to explore and evaluate the visual search behaviour of Arabs and English when searching through the Google Pages. The preliminary results provide new insights into the differences in visual search behaviours and eye movements between members from the Middle East and United Kingdom.

Index Terms—search engines; SERP; eye tracking; visual behaviour; cultural variations; human-computer interaction; usability testing; user experience; human factors.

I. INTRODUCTION

The immense expansion of the Internet makes Web sites in general, seek to attract the users' attention and show their content in an effective way through Web design components such as heading, colours, pictures, and design [1]. The ability to clarify which parts of the web pages receive the most visual attention can benefit both end users and web site author [2]. Search engines are particularly important type of Web site, and recently, the most popular website accessed by global users is search engines¹. In today's life, searching information effectively in internet makes a very big difference in carrying out projects which ranges from academics to business. Search engines are considered as an entrance gate to the internet because it allows to find various pages of interest based on the project needs.

Since search engines attempt to personalise the search results, interpreting and measuring visual search behaviour

of different groups with different cultural and language backgrounds is useful. In addition, enormous practical implications for commercial and academic purposes can be derived from understanding how users explore and search for information in search engine result page. For example global companies can offer and develop strategies for clients in order to increase traffic to their Web sites. Therefore, deep understanding of users visual search behaviour and expectations for search engine from eye tracking can be very beneficial and valuable for companies and their clients [3]. Therefore, it will be interesting to investigate if culture and language characteristics influence the ocular behaviour of the two groups of users when they are searching the SERP.

The main research question that will be addressed is "Do Arabic and English users' differ in their visual search behaviour when using the same English Search Engine?". To address the research question, eye tracking technology will be used on the participants who will be using SERP interface from Google with two different presentations. First one will be plain list of results without images, ads and elements on the right hand side. The second one will be results with images, ads and elements of the right hand side. The reasons for using eye tracking study and SERP are discussed below.

Eye tracking has developed rapidly and becoming a promising tool to answer visual search effectiveness [4]. Recently, the usage of eye tracking technology in online search has received a substantial amount of attention from research scientists, usability experts and marketing companies. In addition, eye tracking is one of the tools that helps to investigate deeply into the users' behaviours during online search. It has been utilized for decades in order to study human behaviours and to contribute to the comprehension of users activities such as scanning, reading and overall processing of visual stimuli [5].

Search Engine Result Page (SERP) is considered as an entrance gate to the internet because it allows to find various pages of interest based on the project needs. The SERP interface from Google was choose as one of the search page as it is the search engine with the highest market share in the world².

¹<http://www.ebizmba.com/articles/most-popular-websites>

The main aim of this study is to explore whether there are cultural variations in visual search behaviour and eye movements on SERP between two different language groups (Arabic and English people), using the eye-tracking method. More precisely, the research examines whether differences of visual search and eye movement exist between the two groups when viewing the same list or ranked results, and if so, whether their eye movements are affected according to the type of task or the type of SERP interface. The significance of this research is to determine if users' cultural and language background differences require local adoption to produce search engine result page to match users' expectations and searching strategies. In addition, comparing two different groups has two different reading directions (English read left-to-right and Arabic read right-to-left) will lead to an interesting exploring study and how users' visual and interacting behaviour relates to cognitive activities for decision making. The outcomes of this research will help the web search engines to display appropriately SERP based on the users' expectations and needs with different cultural background.

As a preliminary work, a controlled eye-tracking experiment was conducted to explore and evaluate the visual search behaviour of Arabs and English when searching through the Google Pages. The preliminary results provide new insights into the differences in visual search behaviours and eye movements between members from the Middle East and the East.

II. BACKGROUND

In order to conduct eye-tracking experiment on SERP, some of the necessary background knowledge is introduced. Three general areas are considered: Cognition, visual behaviour, search engines with different users.

A. Cognition and Web Design

Several researchers argued that cultural context influence the patterns of thinking and process of information [6]. In terms of website design, cultural factors impact on cognitive styles of the designers and can be seen in the production of web sites [6]. A body of research has revealed that culture can shape the individuals' cognitive style which in turn influence on line information retrieving [7]. The study by Chen and Macredie [8] is a representative study that articulate the influence of cognitive style on online learning. For example, when navigating the hypermedia, individuals with different cognitive style showed different learning preferences. Another study was conducted by Graff et al. [7] claimed that the differences in individuals' cognitive style diverge mainly with nationality. They suggested that the design of online aid materials have to take into account the cognitive differences that affected the attitudes. Faiola and MacDorman [9] observed the associations between the cultural differences in designers' cognitive style and how it affects the design of a website. In turn, how this variations influence the preferences and performance of cross-cultural users from different nations. The researchers assumed that the variations in cultural cognitive

style of the designers influence the formula and the structure of the web site. As findings revealed, Americans and Chinese have done the tasks faster and preferred and admired web sites that created by designers from their own culture and whom put into consideration their own cultural aspects. This affirms the significance of designing the web contents and navigations based on users' perspectives to support users' preferences, characteristics and learning styles.

B. Search Engine Result Page (SERP) and Visual Behaviour

Most of the search engine results page SERP contains other items along with query results. It includes Ads in many positions, sponsored links, graphical components like images, maps, etc. The researchers [10][11] conducted an eye tracking experiment to investigate how users distribute their eyes on SERP and which part receive most attention. They found that the top search results received most of the attention and Ads of the top received substantial amount of attention. Moreover, they claimed that the amount of attention depends on the quality of those Ads. Similarly, Guan and Cutrell [12] reported that users gave more attention and spent more time on the top of the SERP while gave less attention and spent less time on the bottom of SERP. As a result, users rely only on the search engines ranking system of web page. Matsuda et al. [13] explored how the position of the search results influence the users' activities. Guan and Cutrell [12] also used eye-tracking method to measure the influence of results rank on users searching. Although the above investigations examined the user's search behaviour in various aspects, to the best of author knowledge, only few references in the literature systemically identify the visual processing of two different groups of people in terms of culture and language. This was the motivation behind the present study.

C. The Influence of Tasks Type and SERP Presentation on Visual Search Behaviour

Users interact and communicate with Web pages through visual elements such as text, images, videos or ads. Moreover, visual behaviour plays an important role in navigating the page. For example, changing the size of elements or changing the image from static to dynamic can influence the visual attentions. Items placed on the top are perceived as more important than placed on the bottom [14]. Similarly, users exhibit tendency to view and process images before other elements [15]. A study conducted by Pan et al. [16] investigated the users' viewing behaviour using eye tracking for 22 different Web pages varying in style, design and type. They induced that the gender, the page order and the type of the page influence viewing behaviours. In a related study that involved users' experience with Google search engine [17] they exposed that users have confidence in the process that Google embraced to show results, and therefore there is a great probability that users often choose the first top results. Most of the search engine present their results in similar format. The rank of the results are considered the main determinant factor affecting users' choice. This was demonstrated by previous

studies such as a study who showed that users preferences are influenced by the results order. Even with synthetic ordering, participants tend to choose results that near to the top. While most of the work described above have investigated different aspects of web search and helped to explain the influence of different kinds of SERP presentations and task performance, few studies have examined how users from different countries respond and interact with SERP in the same manner.

D. Arabic Users and Search Engines

There are only a few studies that dealt with measuring the visual behaviour of Arabic users when using search engines. The majority of studies concerning on how Arabs typing queries and retrieving the information. For example, a study by Aldayel and Ykhlef [18] was done in Saudi Arabia and the results were obtained from online questionnaires of 463 Arabic users. They found that most users do paraphrasing of the queries during search session and they suggest producing like an automated paraphrasing tool, which helps users to obtain desired results. In addition, they confirmed that most users tend to use multilingual search engine for retrieving Arabic content rather using Arabic based search engines. George and Jeyasekhar [19] exhibited the visual reading patterns of Arabic users on Arabic interfaces. They analysed the patterns on e-learning module, newspapers, forums and search engines. Regarding search engines, there were patterns similar to those exhibited from English search engine result page, where users see the title followed by the snippet then the URL of this page. It should be noted from the above literature reviews, however, that limited studies are available on exploring the differences of eye movements on search engines. In particular, Arabic and English users remain unexplored and this was the motivation behind the present study.

III. METHODOLOGY

The only way to record the eye movements of individuals was by eye tracking system. Eye tracking session was considered the main step of data collection and provide the main information for this research. The eye tracking device used is Tobii and Studio software comes along with Tobii for visualizing and analysing the obtain data. To explain in details, a variety of eye movement metrics that will be used in this study which are consistent with the other research papers [16][20]:

- Time to first fixation: the amount of time that participant takes to look at AOI for the first time.
- Total fixation duration: this metric indicates the duration of participant fixation within the specified AOIs and the average fixation duration is an indicator of cognitive processing.
- Fixation count: this measures how many times participants fixated on AOI.
- Heat maps: is used to represent the visual behaviour of participants attention on SERP based on fixations count, where the red spots represents the most fixated and the green and yellow represents the lower fixated. Although

the heat map can be created by using data only obtained from one participant, they are much more useful when aggregating the obtained data from multiple participants.

Area of Interest AOI is an automated tool integrated in Tobii studio software. This tool allows the researcher to divide the displayed stimulus into different regions of interest (e.g. top banner, bottom etc.). The system then tracks participant's eye movement automatically to calculate gaze data through linking the eye gaze with the different areas on the displayed stimuli. This process of identification areas was performed in very early research by Salvucci [21] as a region based method. In this study, eye-tracking software divided each SERPs into Area of Interests (AOIs) that is not visible to the participants. These areas of interests were classified based on common SERP design such as organic results from one to ten, search box, Ads, maps, related search etc.

A. Participants

In total, 36 Arabic and English participants took part in this test and were asked to find a piece of information on SERP. All participants were from Brighton University studying different subjects and pursuing an undergraduate or a postgraduate degree. Each group consisted of 20 participants but due to calibration difficulties, two participants from each group were excluded. Thus, the number became 18 participants for each group were added for analysis. Most of the participants were students and highly educated. All participants are technologically literate and have an experience in using Internet and search engines precisely.

B. Experiment Design

Search Engine Result Page is sometime visually complex, including a variety of design elements. Therefore, the study was designed to show search engine results pages in different configurations. The results pages were retrieved from real Google search for the all tasks and two types of the search engine result page were chosen; one is a plain SERP with no ads, images and enriched snippets, and the other with enriched snippets and elements such as maps, images, related search etc. The purpose of this, trying to cover all kinds of elements that the users might find in different search engines. The study does not aim to explore how certain elements of web page affect the pattern of eye movements but rather it aims to explore how the users actually see the web page that contains different design elements. Therefore, it is more applicable to use real search engine result pages for this study. Both tasks performed were from general domains without specific topics, aiming to reduce the familiarity of those tasks, generalize the findings and trying to counterbalance across participants. At the beginning of each task, a description was presented telling the subjects what they have to look for. In order to ensure equal search results across participant, the experimenter provided the keywords and initial query for each task. In addition, all the potential tasks were submitted earlier to the Google search engine and the relevant SERPs were saved in local files for further modifications. The

other details on how participants answered the questions are discussed in the results section.

IV. PRELIMINARY RESULTS

This section provides the preliminary results from the analysis of the objective data gathered from the first experiment. Specifically, data from the two tasks were analysed only and presented in this section. Most of the data are visualized data and include two groups Arabic and English users. The rest of the findings are demonstrated using charts and tables.

A. Data Analysis and Visualisation

The data from three eye tracking metrics were analysed briefly in this section to introduce an overview of the data obtained from the first experiment. However, the main part of the analysis have not been analysed yet. The three eye tracking metrics that were included in this analysis are time to first fixation, fixations count and total fixation duration. All three metrics were analysed using Excel software and Tobii Studio Software to produce initial descriptive data in the form of tables and charts. Time to first fixation was visualized using a method adopted from Bojko [22].

As this is a preliminary study results, only two tasks are presented; Task A and Task B. Task A was done on the plain SERP and Task B was done on enriched SERP with images and elements on the right hand side.

Task A

As described in the methodology section, 18 participants from each group were asked to find a piece of information on SERP. The task was “Find a book about child care written by Miriam Stoppard” and the answer was in the middle of the results. In the experiment, In the experiment, as shown in Fig. 1, the page was divided into 15 AOIs by using the definition tool to identify where participants are exactly looking to make sense of the data obtained and to make appropriate comparisons for the data. In the following subsections, each metric will be explained separately.

Tasks A	Arabs	Arabs	Arabs	Arabs	English	English	English	English
Area of Interests (AOIs)	Percent of Users Viewed	Number of Fixations Counted	Mean	Percent of Total Fixations	Percent of Users Viewed	Number of Fixations Counted	Mean	Percent of Total Fixations
Top	12%	12	6	6%	0%	0	0	0%
Result1	76%	62	4.76	4%	76%	33	2.53	3%
Result2	88%	136	9.06	8%	94%	88	5.5	6%
Result3	88%	146	9.73	9%	100%	151	8.88	9%
Result4	88%	153	10.2	9%	100%	150	8.82	9%
Result5	100%	213	12.5	11%	100%	159	9.35	10%
Result6	100%	163	9.58	9%	100%	205	12.05	12%
Result7	88%	208	13.86	13%	100%	213	12.52	13%
Result8	65%	137	12.45	11%	71%	131	10.91	11%
Result9	59%	66	6.6	6%	59%	104	10.4	11%
Result10	59%	53	5.3	5%	35%	50	8.33	8%
Result11	53%	53	5.9	5%	35%	53	8.83	9%
Related Search	12%	4	2	2%	0%	0	0	0%
Paging	0%	0	0	0%	0%	0	0	0%
Right Side	6%	1	1	1%	0%	0	0	0%
TOTAL		1407	108.94	100%		1337	98.12	100%

Fig. 1. Fixation data for Task A

1. Time To First Fixation

About 88% of Arabic users fixated first on the Result2 (average time to first fixation for Result2 is 1.18 msec), followed by Top (average time to first fixation for Top 1.85 msec). Almost 100% of English users fixated first on Result3 (with an average

time of 0.23 msec), followed by Result2 (with an average time of 0.48). Interestingly, English users fixated first on Result3, which is the first organic result, as the first two results were advertisements. Although the analysis in this section presents the results only a descriptive level, it is evident that Arabic users took much longer time than English to fixate on the AOIs, where Fig 2. illustrate a visualization of the fixations order for each group on each AOI based on time to first fixation.

2. Fixation Count

The total fixation count for Arabic is 1407 with an average of 108.94 and the total fixation count for English is 1337 with an average of 98.12. The most fixations for Arabic participants were on Result5 followed by Result7 and Result6, whereas English participants fixated mostly on Result7 followed by Result6 and Result5.

Almost 13% of average fixation count for both groups were on Result7, since the answer was in Result7 AOI. Although there was a white space on the right hand side, one fixation was recorded from Arabic participant. This is can not be considered as being of significance as it happened for just one participant with single fixation. Fig. 1 presents the analysed data of Task A for each group. which are; the total fixation count, mean fixation count, percentage of fixation, and the percentage of participants viewing each AOI.

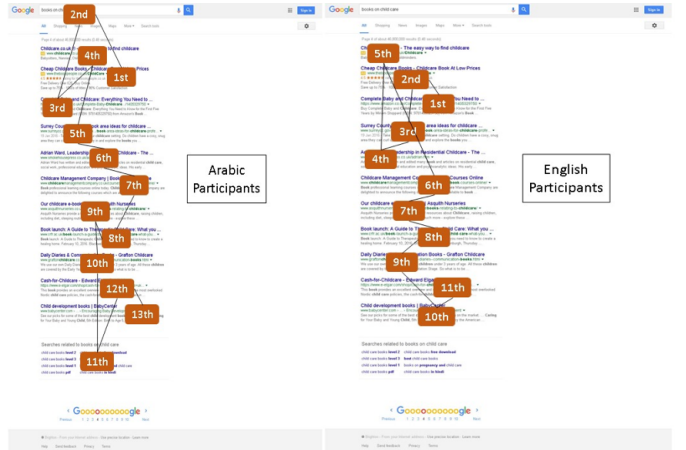


Fig. 2. Fixation order for Arabic and English on each AOIs.

3. Fixation Duration

Total fixation duration for Arabic and English users is 471.3 msec and 372.8. msec respectively. Almost 10% of Arabic participants fixation duration was on the Top and Related Search AOIs, whereas, English participants did not fixate on the TOP neither on Related Search AOIs. The longest fixation duration for both groups were on Result7, as it was the answer for the task A question. In the first three results, Arabic spent 24% of their fixations, while English spent only 17% of their fixation. In the last three results, however, Arabic spent only 13% of their fixations, while English spent more than 28% of their fixations. Fig. 3(a) and Fig. 3(b) are heat maps show the aggregate fixation duration from all Arabic and English

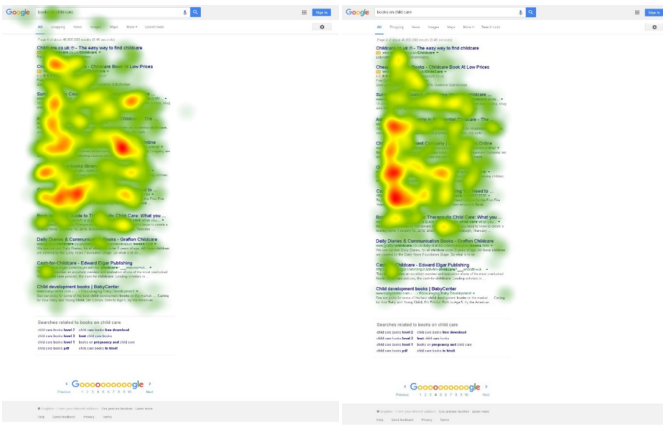


Fig. 3. (a) A heat map for Arabic participants in Task A. (b) A heat map for English participants in Task B.

participants. Red colour indicates the most fixation duration and green colour indicates the lower.

Arabic users tended to read all the results from the beginning of the results until the end, even though the first two results were advertisements, whereas English users tended to jump to the next results without reading the whole sentence and tended to ignore the first two results, as they were advertisements.

Task B

This task was performed on SERP with enriched snippets, images and elements on the right hand side. The participants were required to search the page to find the answer for this question “How many animals in Pretoria Zoo”. The answer was in the middle of the page and the page was divided into 15 AOIs, as shown in Fig. 4.

Tasks A	Arabs	Arabs	Arabs	Arabs	English	English	English	English
Area of Interests (AOIs)	Percent of Users Viewed	Number of Fixations Counted	Mean	Percent of Total Fixations	Percent of Users Viewed	Number of Fixations Counted	Mean	Percent of Total Fixations
Top	12%	5	2.5	3%	0%	0	0	0%
Result1	76%	95	7.3	7%	100%	98	5.76	5%
Result2	76%	77	5.92	6%	94%	86	5.37	5%
Middle Images	88%	127	8.46	9%	88%	95	6.33	5%
Result3	88%	158	10.53	11%	100%	121	7.11	6%
Result4	88%	148	9.86	10%	100%	160	9.41	8%
Result5	59%	61	6.1	6%	76%	96	7.38	6%
Result6	41%	47	6.71	7%	41%	62	8.85	7%
Result7	35%	48	8	8%	24%	82	20.5	17%
Result8	35%	35	5.83	6%	24%	31	7.75	7%
Result9	35%	38	6.33	6%	24%	36	9	8%
Related Search	12%	5	2.5	3%	24%	6	1.5	1%
Paging	6%	2	2	2%	0%	0	0	0%
Top Right	88%	176	11.7	12%	88%	313	20.86	18%
Bottom Right	65%	57	5.18	5%	71%	112	9.33	8%
TOTAL		1079	98.92	100%		1298	119.15	100%

Fig. 4. Fixation data for Task B

1. Time to First Fixation

Arabs fixated first on Result2 and Top as a second fixation same as in Task B. English users fixated first on the Middle Images and then on Result2 as a second fixation. Fig. 5 illustrates the fixation order in the page for each groups. The aim for using SERP with elements on the right hand side was to see whether Arabic users will visit this part faster than English due to their reading direction. However, in this task

English viewed the right hand side faster than Arabic did. Arabic took them an average of 6.56 sec to visit the right hand side and it was the fifth fixation, whereas English participants took them only 1.78 sec and it was the third fixation order.

2. Fixation Count

The total fixation count for Arabic is 1079 with an average of 98.92. The total fixation count for English is 1298 with an average of 119.15. The most fixations for both participants were on the Top Right AOI with an average of 12% for Arabs and 18% for English users. There are two images on the Top Right which might influence the participants and gained more fixations.

The Top AOI in this task also received fixations from Arabs but not from English users as in Task B. Moreover, as shown in Fig. 4 the Top Right and Bottom Right AOIs received much more fixations from English users than Arabs did. Fig. 4 presents the analysed data of Task B for each group which are; the total fixation number, mean fixation count, percentage of fixation, and the percentage of participants viewing each AOI.

3. Fixation Duration

Total fixation duration for Arabic and English participants is 366.9 msec and 351.5 msec respectively. The AOI that has the most fixation time is Top Right AOI for the both groups. 4% of English users fixation duration was on Middle Images, while it almost a double for Arabs with 8%. Result7 received substantial fixation time, as it was the answer for the task. The elements on the right hand side, which represent Top Right and Bottom Right AOIs, Arabic spent only 17% of their fixations, while English spent 27% of their fixation. Fig. 6(a) and Fig. 6(b) are heat maps show the aggregate fixation duration from all Arabic and English participants.

Arabic participants tended to read the first results intensively before moving to the rest of the results listed. Arabs focused on the Middle Images more than English users did. Arabs have more trust on the first three results than bottom results as they fixated more on the top results. English jumped between results without reading the content to pick just the keywords from the snippets to answer the query.

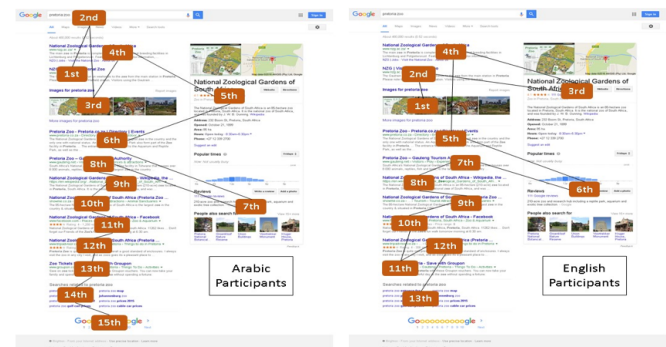


Fig. 5. Fixation order for Arabic and English participants on the page.

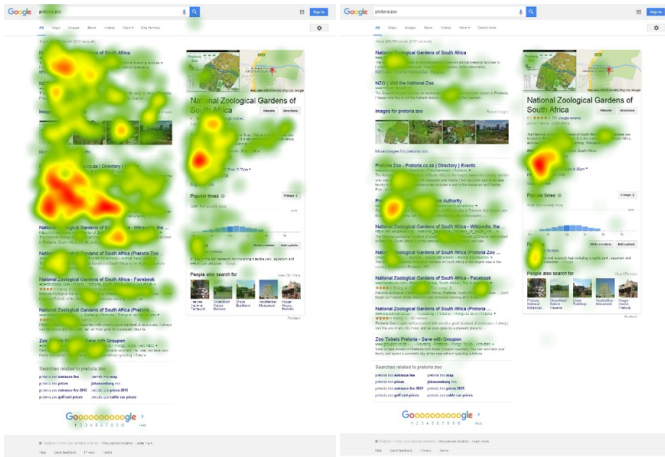


Fig. 6. (a) A heat map for Arabic participants in Task A. (b) A heat map for English participants in Task B.

V. CONCLUSION AND FUTURE DEVELOPMENTS

The initial data that presented in this article were analysed and visualized briefly using Excel software and Tobii visual Studio Software. Descriptive statistical analysis was used concisely to analyse two tasks only. The aim was to give an overview of the data obtained from the first experiment and to clarify the methodology and how far it is applicable to be applied in the rest of this research. The objective data were from three eye tracking metrics (time to first fixation, fixations count and total fixation duration) and the remaining will be included in the final version of this study. The subjective data such as demographic data and satisfaction rating were not included yet due to the limitation of page number. The remaining analyses are considered the main part of this research and it will be done in future work. They contain crucial steps in order to investigate and clarify the differences and similarities between different groups (Arabic and English). The steps include adding Chinese group, use different presentation such as mobile screen, analysing subjective data such as satisfaction rating and examine the influence of task type and SERP type on visual behaviour and performance. This will be done through descriptive and inferential statistical analysis using SPSS and Excel software [23] [24][25]. Moreover, conducting the comparative analyses to find similarities and differences in order to draw solid conclusions about the data gathered. Furthermore, includes mobile phone to the research will enhance the findings and contribute meaningfully to the body of knowledge.

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